

PARTS LIST

Semiconductors:

IC1	- ATmega2560 microcontroller
IC2	- MAX232 RS-232 driver
IC3	- 7805, 5V regulator
D1-D4	- 1N4007 rectifier diode
LED1-LED5	- 5mm LED

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon)

R1-R5	- 470-ohm
R6	- 10-kilo-ohm

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disk
C3, C4	- 22pF ceramic disk
C5	- 4.7 μ F, 16V electrolytic
C6-C10	- 1 μ F, 16V electrolytic

Miscellaneous:

X1	- 230V AC primary to 9V, 500mA secondary transformer
X _{cr}	- 16MHz crystal
S1-S5	- Push-to-on tactile switch
CON3	- 9-pin D-type female connector
CON2	- 6-pin ISP male connector
CON1	- 3-pin berg strip male connector

article into the respective master and slave microcontrollers. The source codes of master and slave demonstrate SPI communication between two microcontrollers (ATmega2560) fitted in two different boards, connected through a six-pin header with pins MOSI, MISO, SCK, RST, VCC and GND. Note that jumper (JP) is not used in synchronous communication between two devices in this example. JP is used only during programming. Switches S1 through S4 on master control LED1 through LED4 of slave board respectively. This represents the communication between two microcontrollers. The six-pin header is also used to program the microcontroller using serial programmer.

Software

The software for microcontroller of master and slave are written in 'C' using the IAR Embedded Workbench integrated development environment. IAR Embedded Workbench is being developed by IAR Systems and ATMEL developers in parallel and hence it generates the optimised code which

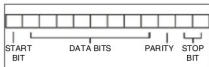


Fig. 5: Asynchronous serial communications sequence

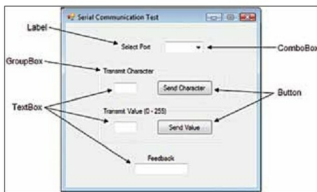


Fig. 6: GUI for serial communication using .Net

uses full 'C' coding capabilities of AVR devices. AVR development tools for embedded systems can be downloaded free from IAR website www.iar.com. For details of IAR Embedded Workbench, you may refer to 'A Beginners' Guide to ATMEL AVR Development' article published in January issue of EBY.

Communication by master. The step-by-step process is as follows:

1. Initialise master
 - (i) Set DDR for MOSI, SCK and $\overline{SS}$ pin as output.
 - (ii) Enable SPI as master by setting SPE and master/slave select (MSTR) bit in SPCR and set clock rate.

2. Master transmits data

- (i) Write data to SPDR
- (ii) Wait while SPIF is set in SPSR
- (iii) May optionally read the data back from slave device

Communication by slave. The step-by-step process is as follows:

1. Initialise slave
 - (i) Set DDR for MISO pin as output
 - (ii) Enable SPI by setting SPE bit in SPCR

(iii) $\overline{SS}$ pin should be driven low by the master

2. Slave receives data

- (i) Wait while SPIF is set in SPSR
- (ii) Read SPDR

Asynchronous serial communications

Universal synchronous and asynchronous receive and transmit (USART) supported by AVR devices is normally an example of asynchronous serial communication, which can also be used for synchronous serial communications. Over the years, several standards have been set for asynchronous serial communications, of which RS-232 standard is most widely used and supported by PC and microcontrollers.

It eliminates the clock signal wire and there

is no cyclic register. So, we are left with only two wires, excluding the common GND wire. MOSI and MISO are represented by T_x and R_x , respectively. R_x and T_x for master and slave devices are cross-connected and each may drive the other one. That is, the T_x pin of master device is connected to R_x pin of slave device and R_x pin of master is connected to T_x pin of the slave device. Each connection sends and receives the data independently. The data-transfer speed is predetermined by a baud rate. The serial data is formatted sequentially as start bit, normally eight data bits – one parity bit and one or two stop bits (refer Fig. 5). The order of transmission is:

1. Each data transmission starts with start bit which is always a zero (0)
2. The data bits are transmitted. The first data bit corresponds to the least significant bit (LSB), while the last bit corresponds to the most significant bit (MSB)
3. The parity bit (if defined) is transmitted
4. Each data transmission ends with stop bit which always has logic level one (1)

To initialise a serial port, we need to activate the functions of the pins R_{xDO} and T_{xDO} for microcontroller ATmega2560. By default, they are enabled in asyn-